

Roshan Hegde

 github.com/RoshanAH  roshanahegde@gmail.com  (952) 594-4175

EDUCATION

University of Illinois at Urbana-Champaign

May 2027

Bachelor of Science in Computer Science, Grainger College of Engineering

GPA: 3.80/4.0

Bachelor of Science in Liberal Arts and Sciences in Mathematics

Courses: Advanced Algorithms, Operating Systems, Programming Languages and Compilers, Abstract Algebra, Graph Theory, Cryptography, Data Structures

EXPERIENCE

W.W. Grainger | *Software Engineer Intern* | *Typescript, PostgreSQL, AWS*

June - August 2026

- Engineered test code migration tool pairing deterministic code transformations and agentic editing to migrate Mocha to Vitest, reducing manual migration work by 80%
- Integrated migration tool across 12 repositories, cutting CI/CD times by 64%
- Designed a closed-loop prompt optimization system using automated benchmarking and feedback scoring, raising agent accuracy from 76% to 90%

Olam Group | *Software Engineer Intern* | *Azure, Python*

May - August 2025

- Analyzed utilization of virtual machines and compute instances using custom scripts and monitoring dashboards, cutting cloud spend by \$30,000 annually
- Built a self-directed data analysis tool to automate VM performance reporting, surfacing over-allocated compute resources for the infrastructure team
- Presented cost-optimization findings to the infrastructure team, informing their resource allocation strategy

UIUC | *Operating Systems Course Assistant* | *C, RISC-V, GDB*

August 2025 - Present

- Developed programming assignments implementing core operating system components – concurrency, virtual memory, and filesystems – on RISC-V QEMU, and rearchitected autograder infrastructure to grade student code concurrently, reducing autograding time from 15 hours to 2 hours
- Hosted office hours for 1000+ students, debugging student implementations of UART drivers, thread schedulers, virtual memory, filesystem drivers, and process isolation

PROJECTS

MechMania | *Lead Developer* | *Go, Rust, Python, Nix*

August 2024 – Present

- Replaced a web-socket-based IPC protocol with a shared memory design using zero-copy serialization and exponential-backoff interprocess polling, resulting in a 1000x speedup over the prior implementation
- Hardened security by using Nix and bubblewrap to sandbox execution of arbitrary participant-written code

Operating System (LumonOS) | *C, Unix, RISC-V, QEMU, GDB*

January – August 2025

- Created fully functional, RISC-V, Unix-based pre-emptive operating system with round robin scheduling policy
- Built a virtio block driver, cryptography driver, and entropy driver
- Implemented support for pipes between processes and argument passing according to POSIX standard
- Placed 3rd in the Operating System design competition hosted by the ECE department

GPT2 Inference Optimization Competition | *1st Place* | *Cuda, C, PTX, Nsight Compute*

December 2025

- Created custom CUDA kernels, profiled with NVIDIA Nsight Compute, achieving a 120x speedup over the baseline and 2x speedup over NVIDIA cuBLAS
- Implemented and tuned optimizations including flash attention, tensor cores, shared memory tiling, and fusion

AWARDS

1st Place, Hackillinois '24, '25 | *Rust, Cargo*

February 2024 and 2025

1st Place, MechMania '24, '25 | *Java, Python*

September 2023 and 2024

SKILLS

Languages: C, C++, Rust, Go, Java, Kotlin, Python, Typescript

Tools: Git, Docker, Linux, Nix, Vim, AWS, Azure, PostgreSQL, GDB